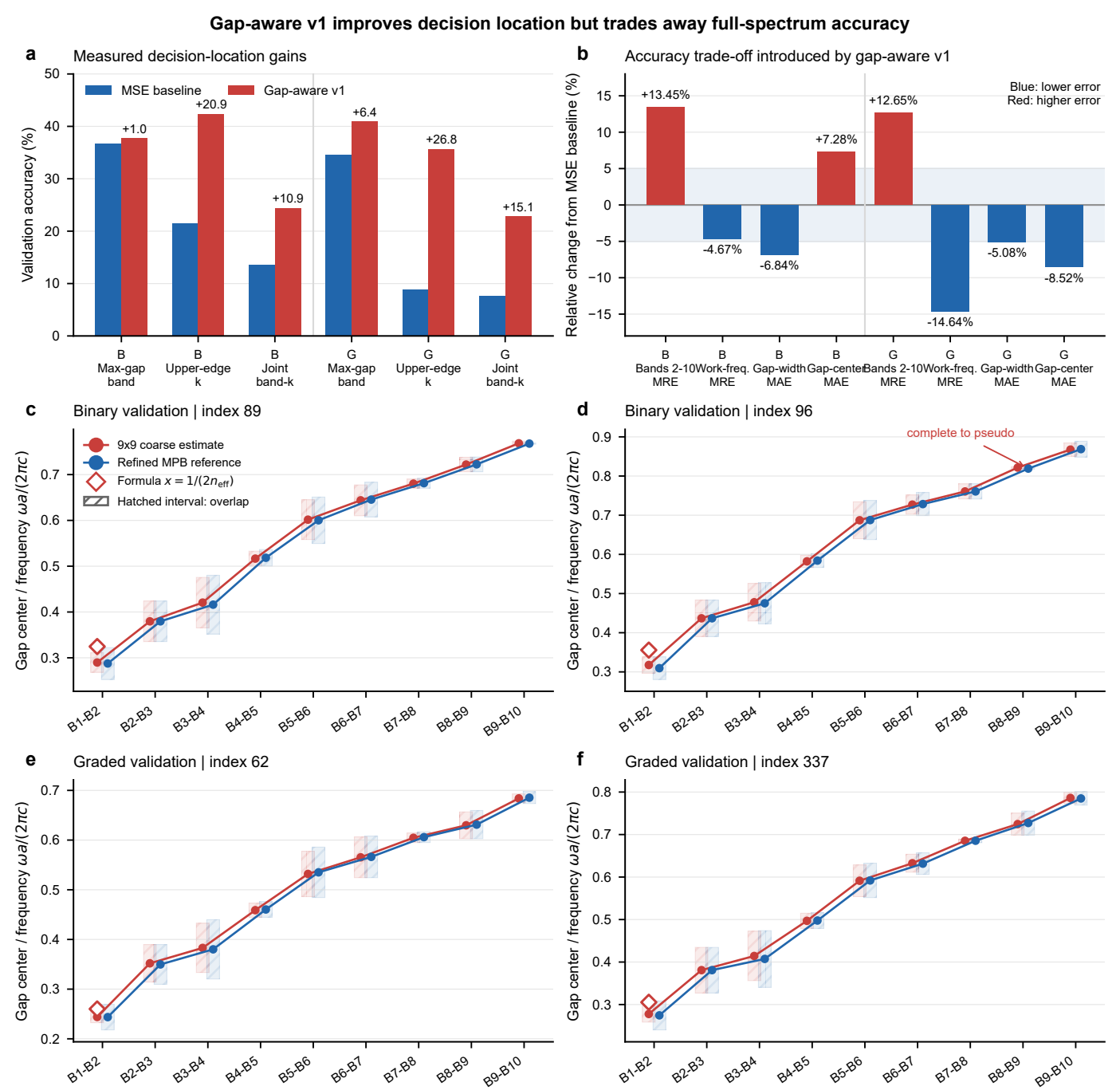

Gap-aware v1 improves decision location but trades away full-spectrum accuracy
a
Measured decision-location gains
MSE baseline
Gap-aware v1
Validation accuracy (%)
+1.0
+20.9
+10.9
+6.4
+26.8
+15.1
B Max-gap band
B Upper-edge k
B Joint band-k
G Max-gap band
G Upper-edge k
G Joint band-k
b
Accuracy trade-off introduced by gap-aware v1
Blue: lower error
Red: higher error
Relative change from MSE baseline (%)
+13.45%
-4.67%
-6.84%
+7.28%
+12.65%
-14.64%
-5.08%
-8.52%
Bands 2-10 MRE
Work-freq. MRE
Gap-width MAE
Gap-center MAE
c
Binary validation | index 89
9x9 coarse estimate
Refined MPB reference
Formula $x = 1/(2n_{\text{eff}})$
Hatched interval: overlap
Gap center / frequency $\omega a/(2\pi c)$
B1-B2
B2-B3
B3-B4
B4-B5
B5-B6
B6-B7
B7-B8
B8-B9
B9-B10
d
Binary validation | index 96
complete to pseudo
e
Graded validation | index 62
f
Graded validation | index 337
a-b: EXP-FWD-014 surrogate validation (seed0, no test). c-f: independent EXP-PHY-005/006 physical-label audit.